Building Industry Management System (BIMS)

Application Programming Interface (API)

DOCUMENTATION

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1. Revision History

Revision Date	Comments	Reviewer
August 2023	Initial Documentation	Denis Ojok

2 Background

BIMS is a web-based application developed by the National Building Review Board (NBRB), with the aim of automating all building control related processes such as application, review, and issuance of building & occupation permits, carrying out routine inspections and appeals as per the mandate of the NBRB.

3 About this API

This is the official BIMS API documentation.

The API is a RESTful API (https://en.wikipedia.org/wiki/Representational_state_transfer) and uses JSON (https://en.wikipedia.org/wiki/JSON) as its data format.

The API utilizes HTTP methods and HTTP status codes to specify requests and responses.

All responses are returned in JSON format.

3.1 API Versioning

The BIMS API uses a semantic versioning scheme (https://en.wikipedia.org/wiki/Software_versioning) to prevent breaking changes.

New API versions are released only when we make backward incompatible changes to the API. We define these as changes that will prevent our official client libraries from working on old versions.

Some changes DO NOT result in a new version, including:

1. Adding new API methods,
2. Adding optional parameters to existing API methods or
3. Adding new fields to responses.

The current version of the API is: v1

4 Using the API

We have production as well as test server environments. The test server environment is called sandbox.

You are advised to test all your API calls on the sandbox environment before pointing your application to production

To point an application to production, you will be required to make only minor changes. In most cases, you will simply need to change the

1. API URL,
2. API TOKEN (where applicable).

4.1 Sandbox Environment

The following is the API Endpoint for our sandbox environment:

https://api.test.bims.go.ug/v1/

4.2 Production Environment

We can't stress this enough: Please test your APIs against our sandbox environment before pointing them to production. We prefer to avoid service disruptions in production caused by untested code.

The following is the API Endpoint for our production environment:

https://api.bims.go.ug/v1/

4.3 Request/Response Format

The default response format is JSON. Requests with a message-body use plain JSON to set or update resource attributes. Successful requests will return a 200 (OK), 201 (CREATED) or a 202 (ACCEPTED) HTTP status.

Some general information about responses:

1. Dates are returned the format: YYYY-MM-DD HH:mm:ss
2. Any decimal monetary amount, such as prices or totals, will be returned as strings with two decimal places.
3. Other amounts, such as item counts, are returned as integers.
4. Blank fields are generally included as null or empty string instead of being omitted.

4.4 Rate Limiting

All requests, whether they are authenticated or not, are subject to rate limiting. If you have reached your limit, your requests will be handled with a **429 Too Many Requests error**. Responses contain several headers which provide information about your current rate limit status.

1. The '**RateLimit-Limit'** header contains the total number of requests you can perform per minute.
2. The '**RateLimit-Remaining'** header contains the number of requests remaining in the current rate limit time frame.
3.

The default limit is 45 requests per second.

4.5 Pagination

Responses which return multiple items support pagination. If they do support pagination, it can be controlled with following query string parameters:

1. A page parameter specifies the page to fetch.
2. The number of the first page is 1.
3. A per_page parameter specifies the number of items returned per page.
4. The default value is 25, the maximum value is 50 except otherwise specified in the documentation.

4.6 Error Handling

BIMS uses HTTP response status codes to indicate success or failure. When a request to our API is successful, The API returns an HTTP response code in the 2XX range. And when a request fails, returns an HTTP response code in the 4XX or 5XX range.

In summary:

1. Codes in the 2XX range mean that the request was processed successfully
2. Codes in the 4XX range mean that something was wrong with the data that you sent. For example, you might have missed some required parameters, or you might be using a wrong API token.
3. Codes in the 5XX range mean that something went wrong within BIMS's servers.

In all cases, the response will include an appropriate status code and content-type. The body of the response will include any additional details regarding the nature of the error.

Most error responses will include a message key in the body of the response, but some may include additional information, for example, they may include information about missing fields.

5 CLIENT API:

All the API endpoints and methods under this module, are designed to be accessed by applications that desire to interact (create, retrieve, update, soft delete) with an applicant/client's data. Examples of such applications may include:

1. Inhouse client facing mobile applications.
2. Third-party Systems interested in verifying the authenticity of building and occupation permits.

5.1 Base URL

The base endpoint for the client module is: **/client**

Which translates to:

https://api.test.bims.go.ug/v1/client on the sandbox environment, and

https://api.bims.go.ug/v1/client on the production environment

5.2 ENDPOINTS

5.2.1 Create Account

Use this endpoint to create/register a new client/user account

Base URL: <API-URL>/client

Method: POST

End point: /account

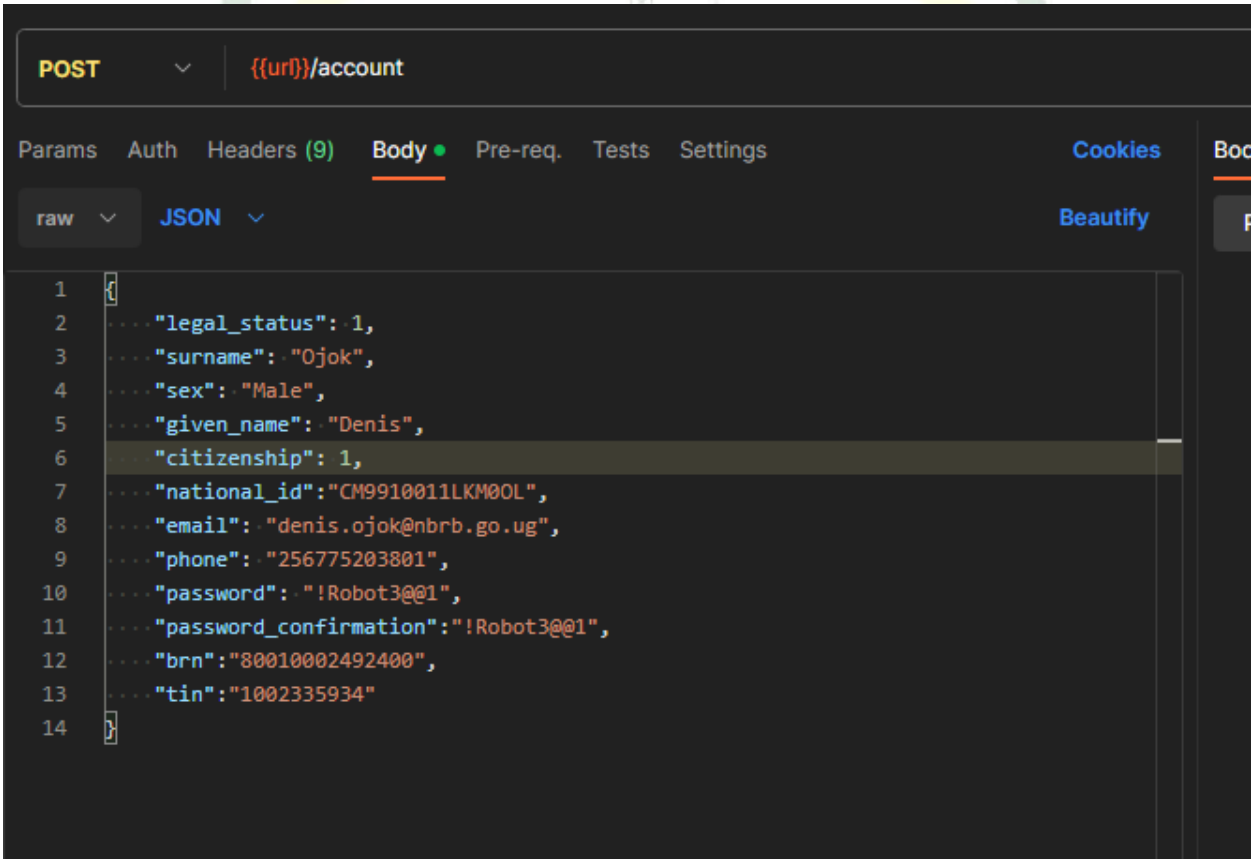
Authentication Token Required: NO

Parameters

#	Parameter	Type	Description
1	**legal_status**	Int (Required)	Legal Status of the applicant and must be either 1, or 2, or 3 as explained below: 1: Individual, 2: Company, 3: Government
2	**surname**	String (Required Only if legal_status is set to 1	Surname of the Applicant
3	**given_name**	String (Required Only if legal_status is set to 1	Given Name of the Applicant
4	**Other_names**	String (Required Only if legal_status is set to 1	Other Name of the Applicant
5	**citizenship**	Int (Required Only if legal_status is set to 1	Citizenship type of the applicant and must be either 1 or 2 1: Ugandan, 2: Non-Ugandan
6	**sex**	String (Required Only if legal_status is set to 1	Sex of the Applicant, Must be either "Male" or "Female"
7	**national_id**	String (Required Only if legal_status is set to 1	NIN Number of the applicant
8	**email**	String (Required)	Email Address of the applicant
9	**phone**	String (Required)	Phone Number of the applicant
10	**password**	String (Required)	Desired Password (set by the applicant)
11	**password_confirmation**	String (Required)	Password Confirmation, Must be the same as 'password'
12	**brn**	String (Required Only if legal_status is set to 2	Business Registration Number of the applicant
13	**tin**	String (Required Only if legal_status is set to 2 or 3	Tax Identification Number of the applicant

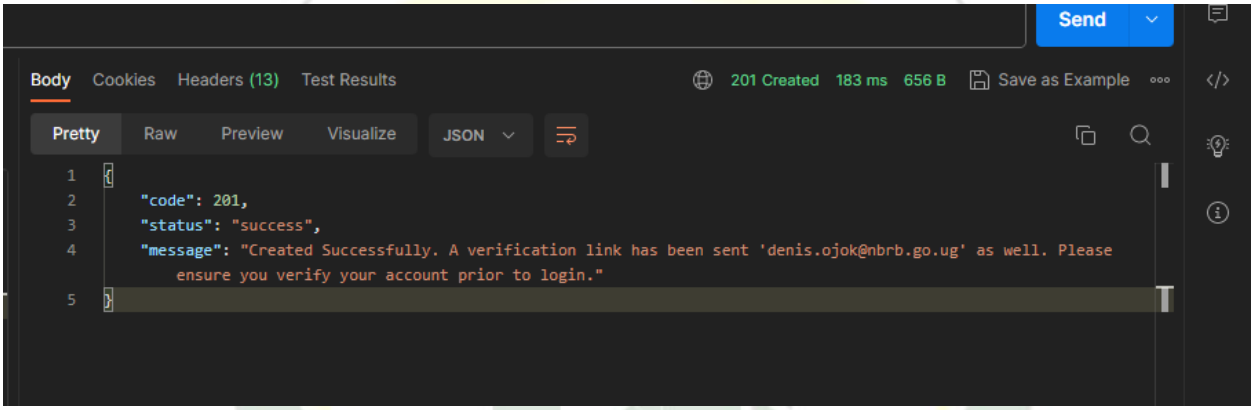
5.2.1.1 Sample Request:

```
curl --location https://api.test.bims.go.ug/v1/client/account' \
--header 'Content-Type: application/json' \
--data-raw '{
    "legal_status": 1,
    "surname": "Ojok",
    "sex": "Male",
    "given_name": "Denis",
    "citizenship": 1,
    "national_id":"CM9910011LKM0OL",
    "email": "denis.ojok@nbrb.go.ug",
    "phone": "256775203801",
    "password": "!Robot3@@1",
    "password_confirmation":"!Robot3@@1",
    "brn":"80010002492400",
    "tin":"1002335934"
}'
```



5.2.1.2 Sample Response:


```
{
    "code": 201,
    "status": "success",
    "message": "Created Successfully. A verification link has been sent 'denis.ojok@nbrb.go.ug' as well. Please ensure you verify your account prior to login."
}
```

5.2.2 Request Token (Login)

Use this endpoint to login a client/user

Base URL: <API-URL>/client

Method: POST

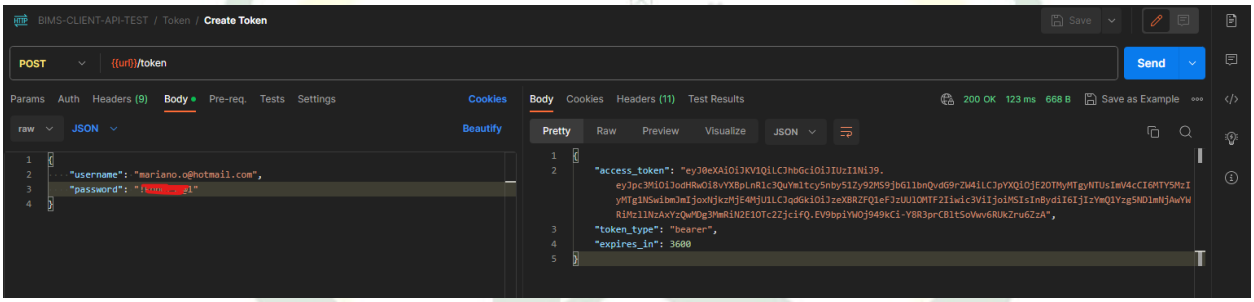
End point: /token

Authentication Token Required: NO

Parameters: username, password

#	Parameter	Type	Description
1	username	String (Required)	An applicant/client's username
2	Password	String (Required)	An applicant/client's password

5.2.2.1 Sample Request & Response:



Note: Upon successful execution, the endpoint above will return an access_token as per the above screenshot, this is the access_token that you will use to authenticate your next requests, by passing it in the "Authorization: bearer" header of each of your requests.

Please note that for security reasons, the access_token will be valid for a maximum of 3600 seconds (1 Hour), You may refresh the token (using the endpoint below) prior or after its expiration.

5.2.3 Refresh Token (Obtain another token)

Use this endpoint to refresh an active token.

Base URL: <API-URL>/client

Method: POST

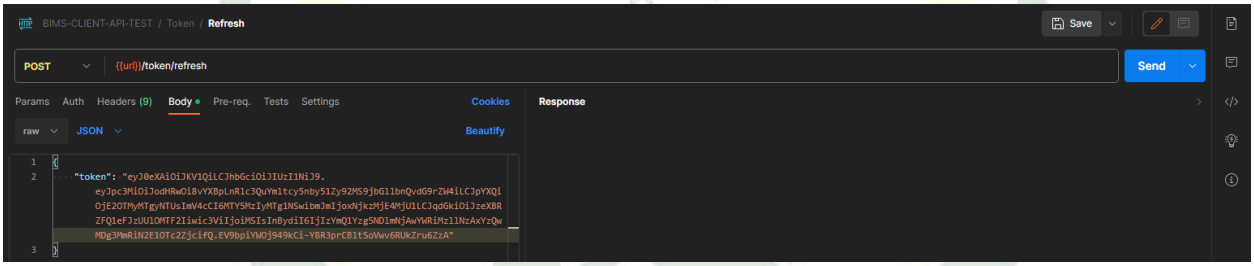
End point: /token/refresh

Authentication Token Required: NO

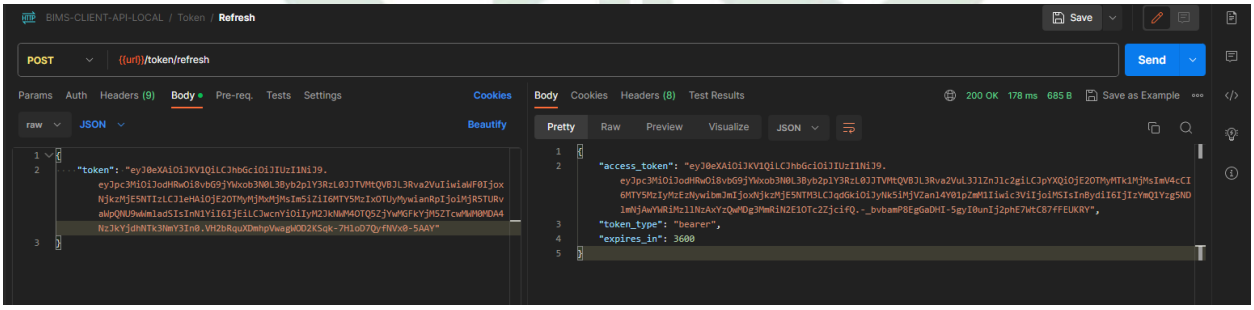
Parameters: token

#	Parameter	Type	Description
1	token	Json	An earlier generated token

5.2.3.1 Sample Request:



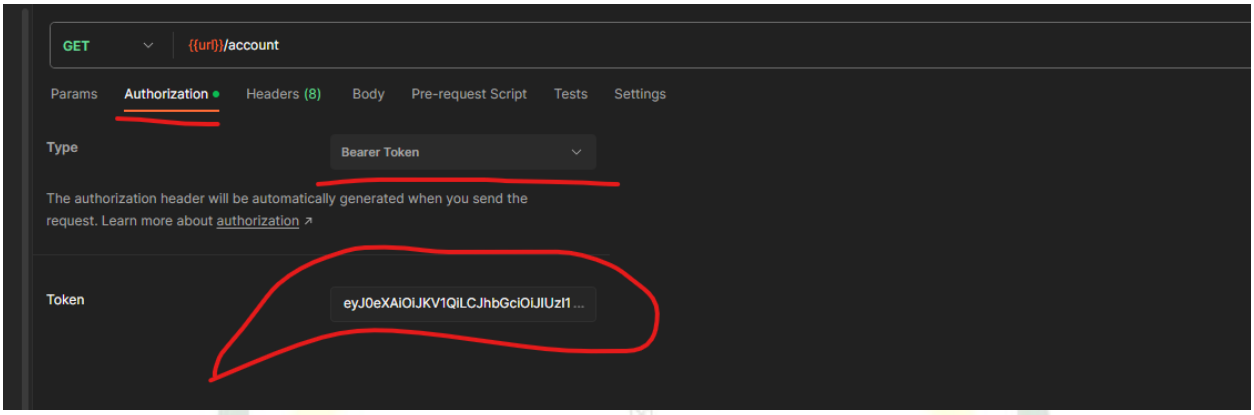
5.2.3.2 Sample Response:



5.2.4 API Authentication

From this point onwards (with an exception of the verifyPermit endpoint), All requests to the API must be authenticated via an API Access token.

Include your API Access token (The one obtained earlier using the /token endpoints) in every request you send to the API, and this must be included with in the Authorization HTTP header.



5.2.5 Get Account Information

Use this endpoint to retrieve a client's profile / Account Details.

Base URL: <API-URL>/client

Method: GET

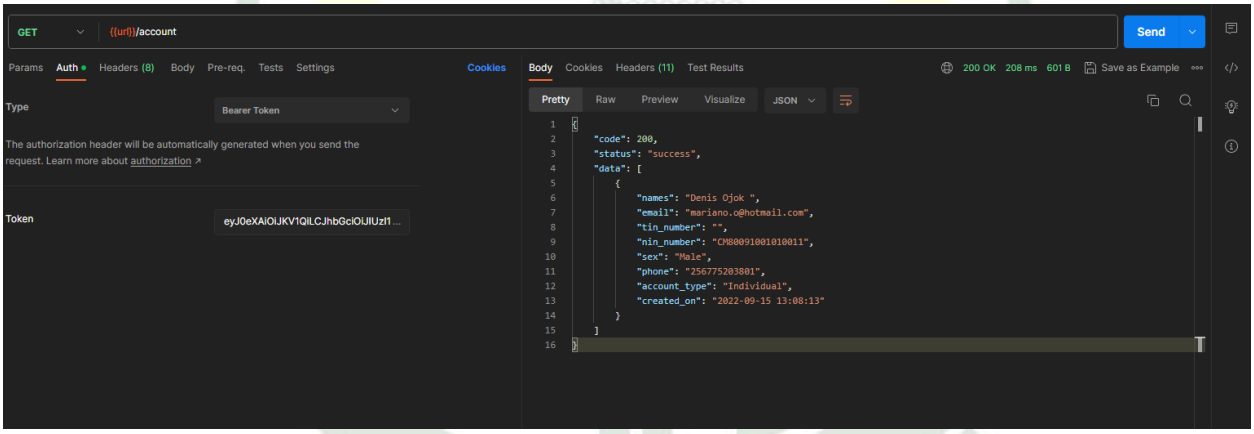
End point: /account

Authentication Token Required: YES

Parameters: None

#	Parameter	Description
-	-	-

5.2.5.1 Sample Request & Response:



5.2.6 Update Account Information

Use this endpoint to update / make amendments to a client's profile / Account Details.

Base URL: **<API-URL>/client**

Method: **PUT**

End point: **/account**

Authentication Token Required: **YES**

Parameters

#	Parameter	Type	Description
1	**legal_status**	Int (Required)	Legal Status of the applicant and must be either 1, or 2, or 3 as explained below: 1: Individual, 2: Company, 3: Government
2	**surname**	String (Required Only if legal_status is set to 1	Surname of the Applicant
3	**given_name**	String (Required Only if legal_status is set to 1	Given Name of the Applicant
4	**Other_names**	String (Required Only if legal_status is set to 1	Other Name of the Applicant
5	**email**	String (Required)	Email Address of the applicant
9	**phone**	String (Required)	Phone Number of the applicant
6	**password**	String (Required)	Desired Password (set by the applicant)
7	**password_confirmation**	String (Required)	Password Confirmation, Must be the same as 'password'

5.2.7 Get All Applications

Use this endpoint to retrieve a list of all open applications belonging to the authenticated user.

Base URL: <API-URL>/client

Method: GET

End point: /applications

Authentication Token Required: YES

Mandatory Parameters: None

Additional Parameters (Optional):

#	Parameter	Type	Description
1	**start**	String/Datetime (Optional)	This is of the format YYYY-MM-DD HH:mm:ss, e.g 2020-05-15 18:44:44 – Useful for filtering applications within a given date range
2	**end**	String/Datetime (Optional)	This is of the format YYYY-MM-DD HH:mm:ss, e.g 2020-05-15 18:44:44– Useful for filtering applications within a given date range
	status	String (Optional)	Filter by application status. eg PENDING PROFFESSIONAL ACKNOWLEDGEMENT, PENDING PAYMENT
	per_page	Int (Optional)	Total Number of records to return. If not specified, the 25 records will be returned by default

5.2.7.1 Sample Request & Response:

BIMS-CLIENT-API-TEST / Applications / **GetApplications**

Save

GET {{url}}/applications

Send

Params Auth Headers (10) Body Pre-req. Tests Settings Cookies

raw JSON Beautify

```
{
    "per_page": 20,
    "start":"2023-01-01 00:00:00",
    "end":"2023-12-31 23:59:59",
}
```

Body Cookies Headers (11) Test Results 200 OK 51 ms 1.15 KB Save as Example

Pretty Raw Preview Visualize JSON


```
{
    "code": 200,
    "status": "success",
    "data": {
        "current_page": 1,
        "data": [
            {
                "application_key": "2209229180",
                "application_type": "Building Permit Application",
                "building_classification": "Class A",
                "building_operation": "Erection of new building",
                "building_purpose": "Arcade",
                "administrative_unit_type": "City",
                "administrative_unit_name": "Kampala",
                "location": "Kirinya",
                "total_square_metres": 2400,
                "created_on": "2022-09-22 12:21:21",
                "update_on": "2023-02-28 06:36:05"
            },
            {
                "application_key": "2209222308",
                "application_type": "Building Permit Application",
                "building_classification": "Class B",
                "building_operation": "Erection of new building",
                "building_purpose": "Residential",
                "administrative_unit_type": "City",
                "administrative_unit_name": "Jinja",
                "location": "Kirinya",
                "total_square_metres": 1600,
                "created_on": "2022-09-22 12:45:11",
                "update_on": "2022-09-22 12:50:43"
            },
            {
                "application_key": "2211176981",
                "application_type": "Building Permit Application",
                "building_classification": "Class B",
                "building_operation": "Erection of new building",
                "building_purpose": "Residential",
                "administrative_unit_type": "City",
                "administrative_unit_name": "Masaka",
                "location": "Nyendo",
                "total_square_metres": 676,
                "created_on": "2022-11-17 08:33:42",
                "update_on": "2022-11-17 09:10:06"
            },
            {
                "application_key": "2211282401",
```


Runner Capture requests Cookies Trash

5.2.8 Get Details of an Application

Use this endpoint to retrieve details of a particular application.

Base URL: <API-URL>/client

Method: GET

End point: /applications/<APPLICATION_KEY>

Authentication Token Required: YES

Mandatory Parameters: Application Key (Must be passed in the URL)

Additional Parameters: None

#	Parameter	Description
-	-	-

5.2.8.1 Sample Request & Response:

GET {{url}}/applications/2209229180 Send

Params Auth Headers (8) Body Pre-req. Tests Settings Cookies

none

This request does not have a body

Body Cookies Headers (11) Test Results 200 OK 24.48 s 882 B Save as Example

Pretty Raw Preview Visualize JSON


```
{
    "code": 200,
    "status": "success",
    "data": {
        "application_key": "2209229180",
        "application_type": "Building Permit Application",
        "building_classification": "Class A",
        "building_operation": "Erection of new building",
        "building_purpose": "Arcade",
        "administrative_unit_type": "City",
        "administrative_unit_name": "Kampala",
        "total_square_metres": 2400,
        "created": "2022-09-22 12:21:21",
        "updated": "2023-02-28 06:36:05",
        "expires": null,
        "status": null,
        "applicant": {
            "name": "Denis Ojok ",
            "phone": "256775203801",
            "email": "mariano.o@hotmail.com",
            "tin": "",
            "nin": "CM80091001010011"
        },
        "professionals_engaged": {
            "architect": {
                "reg_no": "272",
                "name": "Jasilwanyi Olowo Jerome Stowell",
                "address": "National Building Review Board"
            },
            "structural_engineer": {
                "reg_no": "1330",
                "name": "Ssali James",
                "address": "P.O Box 27570, Kampala",
                "discipline": "Civil"
            },
            "quantity_surveyor": {
                "reg_no": "SRB375",
                "name": "Hudson Mutalya",
                "address": "National Building Review Board"
            },
            "services_engineer_mechanical": [],
            "services_engineer_electrical": []
        },
        "audit_trail": []
    }
}
```

5.2.9 **Get All Invoices**

Use this endpoint to retrieve a list of all invoices generated by a client.

Base URL: <API-URL>/client

Method: GET

End point: /invoices

Authentication Token Required: YES

Mandatory Parameters: None

#	Parameter	Type	Description
1	**start**	String/Datetime (Optional)	This is of the format YYYY-MM-DD HH:mm:ss, e.g 2020-05-15 18:44:44 – Useful for filtering invoices within a given date range
2	**end**	String/Datetime (Optional)	This is of the format YYYY-MM-DD HH:mm:ss, e.g 2020-05-15 18:44:44– Useful for filtering invoices within a given date range
	status	String (Optional)	Filter by invoice status. eg PENDING, PAID
	per_page	Int (Optional)	Total Number of records to return. If not specified, the 25 records will be returned by default

5.2.9.1 Sample Request & Response:

GET {{url}}/invoices

Params Auth Headers (8) Body Pre-req. Tests Settings Cookies

Type Bearer Token

The authorization header will be automatically generated when you send the request. Learn more about authorization ↗

Token eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1 ...

Body Cookies Headers (11) Test Results 200 OK 47 ms 1.09 KB

Pretty Raw Preview Visualize JSON


```
{
    "code": 200,
    "status": "success",
    "invoices": {
        "current_page": 1,
        "data": [
            {
                "application_key": "2209229180",
                "prn": "2230000005509",
                "search_code": "SGG4FCLHRQFL",
                "declaration_type": "Original Assessment",
                "assessment_amount": "7680000.00",
                "inspection_fees": "2400000.00",
                "landscaping_fees": "0.00",
                "payment_status": true,
                "created": "2022-09-22 12:35:32",
                "expires": "13-OCT-22"
            },
            {
                "application_key": "2209222308",
                "prn": "2230000005512",
                "search_code": "SGG4FCLK6F45",
                "declaration_type": "Original Assessment",
                "assessment_amount": "4320000.00",
                "inspection_fees": "1600000.00",
                "landscaping_fees": "0.00",
                "payment_status": true,
                "created": "2022-09-22 12:50:03",
                "expires": "13-OCT-22"
            },
            {
                "application_key": "2211176981",
                "prn": "2230000017454",
                "search_code": "SGG4FLTA0EF7",
                "declaration_type": "Original Assessment",
                "assessment_amount": "1825200.00",
                "inspection_fees": "676000.00",
                "landscaping_fees": "0.00",
                "payment_status": true,
                "created": "2022-11-17 09:01:49",
                "expires": "08-DEC-22"
            },
            {
                "application_key": "2211282401",
                "prn": "2230000022943",
                "search_code": "SGG4FQ1RRSAM",
                "declaration_type": "Original Assessment",
```


Runner Capture reques

5.2.10 Get Details of a Particular Invoice

Use this endpoint to retrieve details of a particular Invoice.

Base URL: <API-URL>/client

Method: GET

End point: /invoices/<PRN>

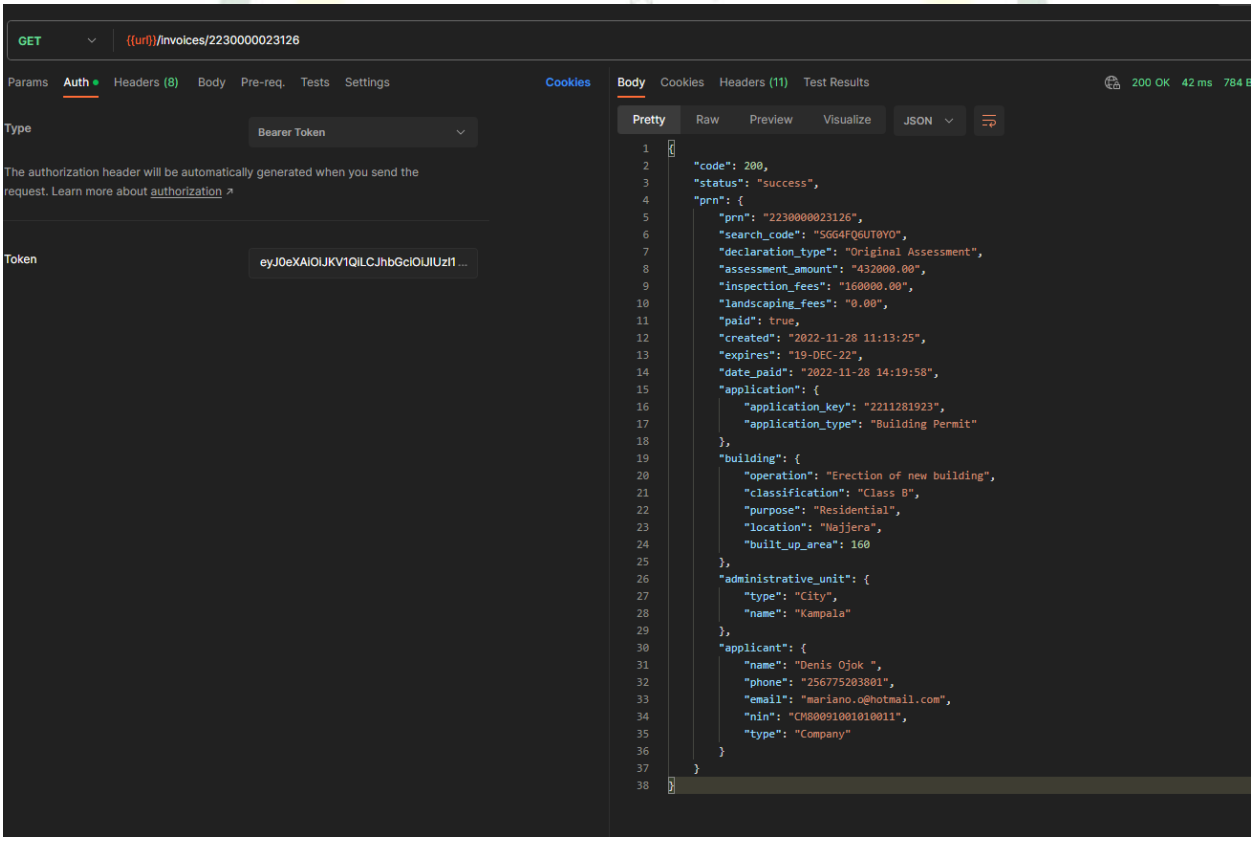
Authentication Token Required: YES

Mandatory Parameters: Payment Registration Number (PRN)

Additional Parameters: None

#	Parameter	Description
-	-	-

5.2.10.1 Sample Request & Response:



5.2.11 Get Details of an Approved Permit

Use this endpoint to retrieve a list of all approved permits belonging to the authenticated user.

Base URL: <API-URL>/client

Method: GET

End point: /permits/<PERMIT_SERIAL_NO>

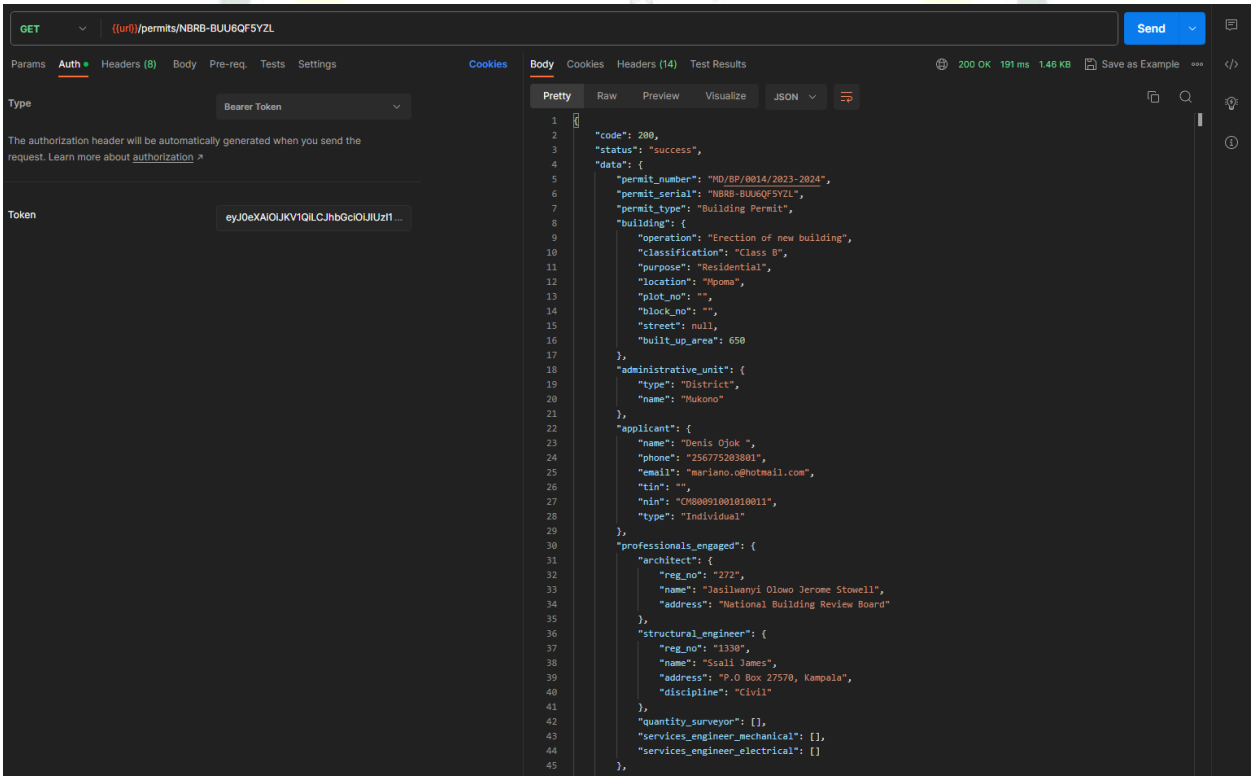
Authentication Token Required: YES

Mandatory Parameters: None

Additional Parameters: None

#	Parameter	Description
-	-	-

5.2.11.1 Sample Request & Response:



5.2.12 Get Approved Permits

Use this endpoint to retrieve a list of all approved permits belonging to the authenticated user.

Base URL: <API-URL>/client

Method: GET

End point: /permits

Authentication Token Required: YES

Mandatory Parameters: None

Additional Parameters: None

#	Parameter	Type	Description
1	**start**	String/Datetime (Optional)	This is of the format YYYY-MM-DD HH:mm:ss, e.g 2020-05-15 18:44:44 – Useful for filtering permits within a given date range
2	**end**	String/Datetime (Optional)	This is of the format YYYY-MM-DD HH:mm:ss, e.g 2020-05-15 18:44:44– Useful for filtering permits within a given date range
	status	String (Optional)	Filter by application status. eg Active, Expired, Cancelled
	per_page	Int (Optional)	Total Number of records to return. If not specified, the 25 records will be returned by default

5.2.12.1 Sample Request & Response:

GET {{url}}/permits

Send

Params Auth Headers (8) Body Pre-req. Tests Settings Cookies

Type Bearer Token

The authorization header will be automatically generated when you send the request. Learn more about authorization ↗

Token eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1 ...

Body Cookies Headers (11) Test Results 200 OK 44 ms 828 B Save as Example

Pretty Raw Preview Visualize JSON


```
{
    "code": 200,
    "status": "success",
    "permits": {
        "current_page": 1,
        "data": [
            {
                "permit_serial": "NBRB-BUU6QF5YZL",
                "permit_number": "MD/BP/0014/2023-2024",
                "permit_type": "Building Permit",
                "building_classification": "Class B",
                "administrative_unit_type": "District",
                "administrative_unit_name": "Mukono",
                "location": "Mpoma",
                "permit_issue_date": "2023-01-18 12:06:51",
                "permit_expiry_date": "2028-01-18 12:06:51"
            },
            {
                "permit_serial": "NBRB-4WUZ3NU45O",
                "permit_number": "LC/BP/00002/2023",
                "permit_type": "Building Permit",
                "building_classification": "Class B",
                "administrative_unit_type": "City",
                "administrative_unit_name": "Lira",
                "location": "Obote Avenue",
                "permit_issue_date": "2023-03-22 14:22:46",
                "permit_expiry_date": "2028-03-22 14:22:46"
            }
        ],
        "first_page_url": "http://api.test.bims.go.ug/v1/client/permits?page=1",
        "from": 1,
        "last_page": 1,
        "last_page_url": "http://api.test.bims.go.ug/v1/client/permits?page=1",
        "links": [
            {
                "url": null,
                "label": "&laquo; Previous",
                "active": false
            },
            {
                "url": "http://api.test.bims.go.ug/v1/client/permits?page=1",
                "label": "1",
                "active": true
            },
            {
                "url": null,
                "label": "Next &raquo;",
```


Runner Capture requests Cookies Trash

5.2.13 Get Details of an Approved Permit

Use this endpoint to retrieve a list of all approved permits belonging to the authenticated user.

Base URL: <API-URL>/client

Method: GET

End point: /permits/<PERMIT_SERIAL_NO>

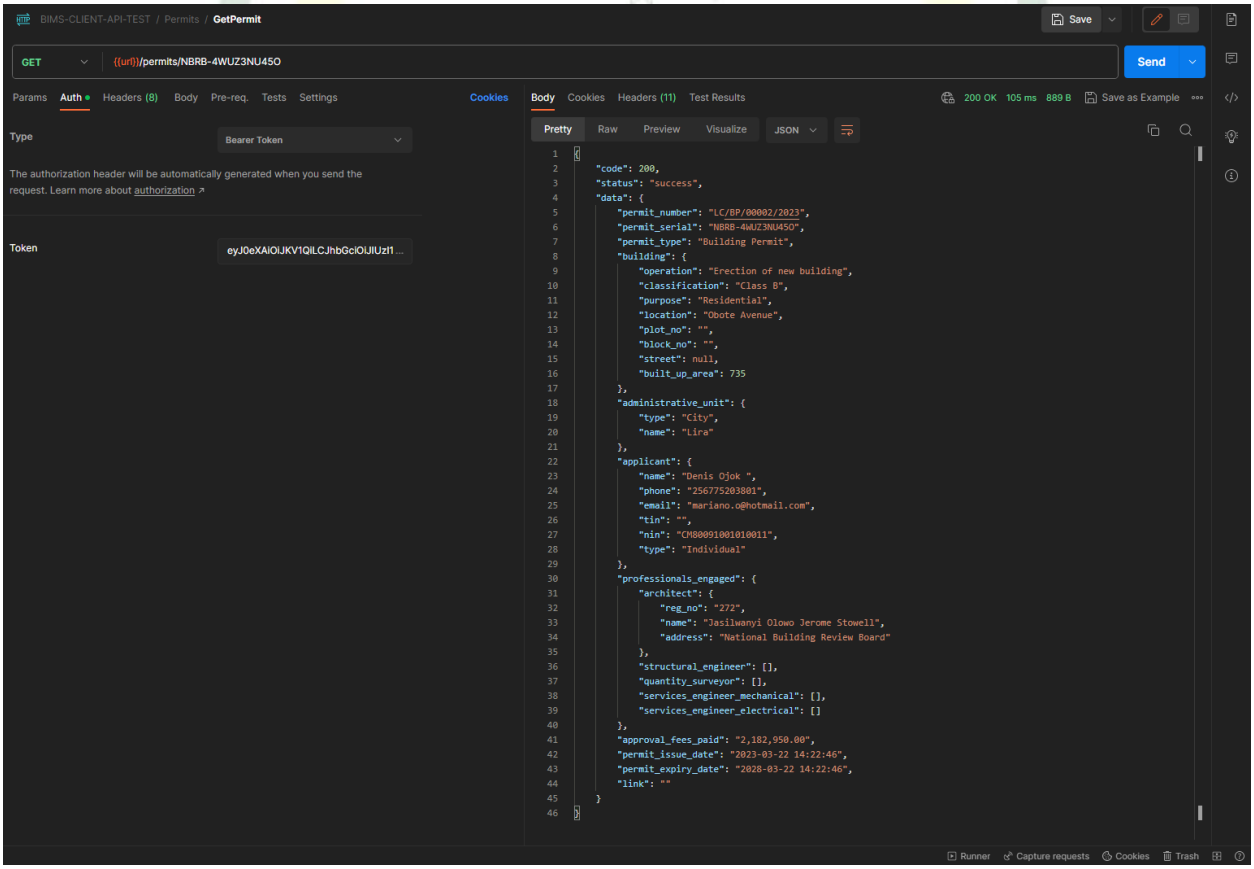
Authentication Token Required: YES

Mandatory Parameters: None

Additional Parameters: None

#	Parameter	Description
-	-	-

5.2.13.1 Sample Request & Response:



5.2.14 Verify Permit

Use this endpoint verify a permit.

Base URL: <API-URL>/client

Method: GET

End point: /permits/verify/<PERMIT_SERIAL_NO>

Authentication Token Required: NO

Mandatory Parameters: None

Additional Parameters: None

#	Parameter	Description
-	-	-

5.2.14.1 Sample Request & Response:

```
{
    "code": 200,
    "status": "success",
    "data": {
        "permit_number": "LC/BP/00002/2023",
        "permit_serial": "NBRB-4WUZ3NU45O",
        "permit_type": "Building Permit",
        "building": {
            "operation": "Erection of new building",
            "classification": "Class B",
            "purpose": "Residential",
            "location": "Obote Avenue",
            "plot_no": "",
            "block_no": "",
            "street": null,
            "built_up_area": 735
        },
        "administrative_unit": {
            "type": "City",
            "name": "Lira"
        },
        "applicant": {
            "name": "Denis Ojok ",
            "phone": "256775203801",
            "email": "mariano.o@hotmail.com",
            "tin": "",
            "nin": "CM80091001010011",
            "type": "Individual"
        },
        "professionals_engaged": {
            "architect": {
                "reg_no": "272",
                "name": "Jasilwanyi Olowo Jerome Stowell",
                "address": "National Building Review Board"
            },
            "structural_engineer": [],
            "quantity_surveyor": [],
            "services_engineer_mechanical": [],
            "services_engineer_electrical": []
        },
        "approval_fees_paid": "2,182,950.00",
        "permit_issue_date": "2023-03-22 14:22:46",
        "permit_expiry_date": "2028-03-22 14:22:46",
        "link": ""
    }
}
```